

YVES ALAIN IRAGENA

Full-Stack AI Engineer · Washington D.C. · Available immediately (F-1 CPT)
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Full-Stack AI Engineer and **ICME 2026 Grand Challenge contributor** building agentic systems, multimodal AI, and LLM infrastructure. Ships complete products: autonomous agents that solve real business KPIs, adversarial RAG systems with **82% hallucination reduction**, and research-grade deepfake detection at **0.8923 macro-F1**. Graduate Research Assistant, MAIL Lab — Catholic University of America.

EXPERIENCE

Graduate Research Assistant · MAIL Lab, Catholic University of America · Washington D.C.

Jan 2025 – Present

- **Achieved 0.8923 macro-F1** on audio deepfake detection (CompSpoonV2, 250K clips, 5 classes) building **DPADD-ES** — mask-based architecture separating speech and environment representations via Frobenius orthogonality loss; submitted to **ICME 2026 Grand Challenge (ESDD2)**.
- **Reduced hallucinations by 82%** building an adversarial RAG security framework with prompt shielding, semantic re-ranking, and output validation across Pinecone, FAISS, and Chroma.
- Deploying real-time synthetic speech classifier (Wav2Vec2 + BiLSTM + attention) to MAIL Lab's internal QA pipeline — validates 500+ audio submissions/day with inference latency under 200ms.

AI & Data Engineer · Health Target Parapharmacy Ltd · Kigali, Rwanda

Oct 2023 – Jan 2025

- Engineered Python/SQL ETL pipelines processing **10K+ monthly transactions** across a 12K-SKU product database — automated anomaly detection eliminated 20+ hrs/week of manual work and achieved a **70% reduction** in data entry errors across 4 retail locations.
- Built inventory intelligence system using pandas + rule-based forecasting across wholesale and import divisions — reduced overstock carrying costs by ~30%, recovering an estimated **\$40K/yr** in tied-up working capital and eliminating fulfillment delays.

TECHNICAL SKILLS

AI / ML	PyTorch · Wav2Vec2 · BiLSTM · Multi-Head Attention · CNN · Scikit-learn · HuggingFace
LLM / Agents	MCP · LangGraph · LangChain · RAG · Pinecone · FAISS · Chroma · Claude API · OpenAI API
Backend / Infra	Python · FastAPI · Node.js · PostgreSQL · Docker · Kubernetes · AWS · GitHub Actions
Frontend	React · Next.js · TypeScript · WebSockets · Tailwind CSS · Cursor · Claude Code

SELECTED PROJECTS

AuraCommerce — Agentic BI Dashboard

[GitHub](#) [Live](#)

Python · FastAPI · OpenAI GPT-4o-mini · SQLite · Plotly.js · Prometheus

3-agent pipeline (Analyst → Sentinel → Reporter) monitoring e-commerce KPIs in real time. Dual-mode Sentinel: GPT-4o-mini + rule-based fallback for zero-downtime reliability.

Gmail AI Semantic Autopilot

[GitHub](#) [Live](#)

Python · Claude Haiku · Anthropic SDK · MCP · Gmail API

Daemon classifying emails into 7 intent categories — 98%+ accuracy, auto-archives cold outreach and newsletters, saves ~10 hrs/week at <\$0.01/day API cost.

DPADD-ES — Audio Deepfake Detection

[GitHub](#) [Live](#)

PyTorch · Wav2Vec2 · BiLSTM · Attention · CompSpoonV2

Mask separation model isolating speech artifacts from background — 0.8923 macro-F1 vs 0.63 baseline on unseen generators. ICME 2026 Grand Challenge (ESDD2).

Real-Time Ops Dashboard

[GitHub](#) [Live](#)

Next.js 16 · TypeScript · Supabase Realtime · Tailwind CSS 4

Live task board on Supabase postgres_changes subscriptions — DB writes on screen in <50ms. Optimistic UI with automatic rollback on write failure.

JP AutoHub SEO Engine

[GitHub](#) [Live](#)

Python · LangChain · AWS · PostgreSQL

Multi-agent LangChain pipeline for SEO automation — keyword extraction, AI content generation, and publishing on a live auto-parts e-commerce platform.

EDUCATION

M.S. Data Analytics — Multimodal AI Focus · Catholic University of America, Washington D.C. Expected July 2027 GPA: 3.7/4.0 · RA · F-1 CPT

Available immediately for internship/co-op — no sponsorship required.